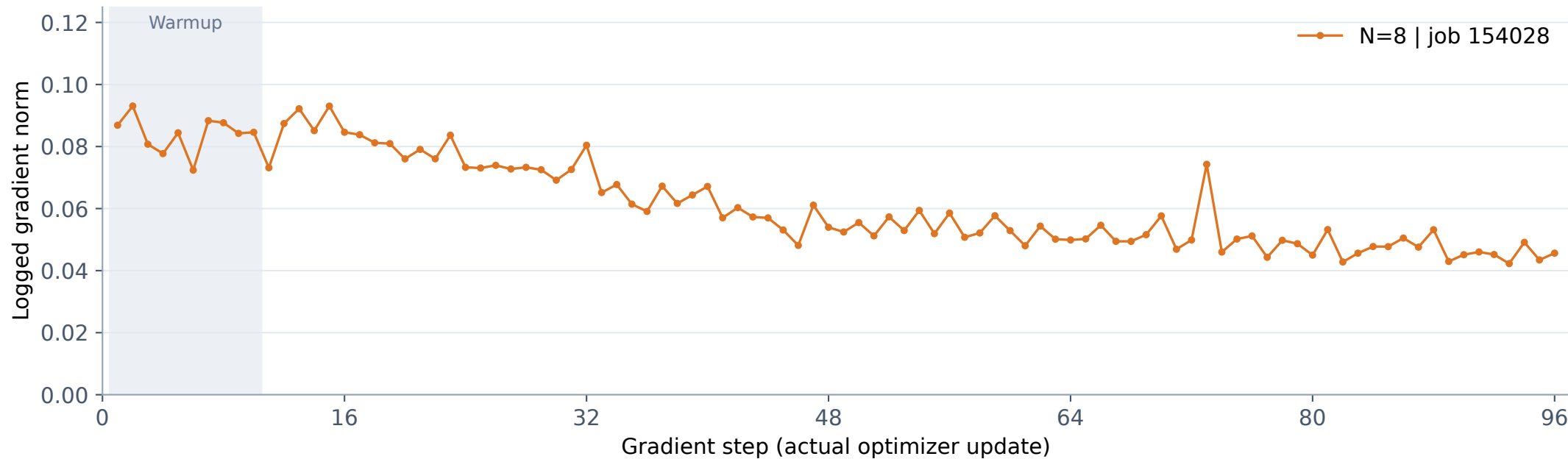


Gradient norm across training | N=8



One point per actual optimizer update; raw values, no smoothing.

N=4 and N=8 use the same axis ranges; shading marks the first 10 warmup updates.